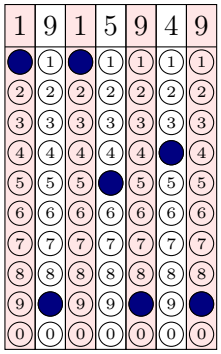
TEST VARIANTI RAQAMI: **1915949**

MATEMATIKA

Ushbu test varianti 30 ta test topshirig‘idan iborat. Uni bajarishga 1 soat vaqt belgilash tavsiya etiladi. Natijani bilish uchun javoblaringizni javoblar varaqasiga belgilab, rasmini **@dtmxizmatbot** orqali jo‘nating. Sizga omad tilaymiz!

TEST VARIANTI RAQAMINI JAVOBLAR VARAQASIGA TO‘G‘RI KO‘CHIRING!



1. $f(x) = x^2 - x + 1$ funksiyaning $(0;\ 2)$ nuqtadan o‘tuvchi boshlang‘ich funksiyasini toping.

A) $F(x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - 1$

B) $F(x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + 2$

C) $F(x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + 1$

D) $F(x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - 2$

2. $3\,(4x - 3y) - 2\,(-3y + 4x)$ ifodani soddalashtiring.

A) $4x - 3y$ B) $-4x + 3y$ C) $-4x - 3y$ D) $4x + 3y$

3. Hisoblang: $\dfrac{(27 + 79) \cdot \left(2 + \dfrac{16}{45}\right) \cdot 45^{-1}}{\left(0,(55) + \dfrac{1}{0,(555)}\right)^2} \cdot 0,(5)$

A) $1\frac{4}{5}$ B) 9 C) 1 D) $0,(5)$

4. Ketma-ket kelgan uchta tub sonlar yig‘indisi quyidagi sonlardan qaysi biriga teng bo‘lishi mumkin?

A) *9* B) *6* C) *21* D) *15*

5. $\bar{a}\,(12; -5)$ vektor bilan Ox o‘qi orasidagi burchak kosinusini toping.

A) $-\frac{5}{13}$ B) $-\frac{12}{5}$ C) $\frac{12}{13}$ D) $-\frac{5}{12}$

6. $\left(4\frac{3}{8} - 1\frac{3}{4}\right) : 1\frac{3}{4}$ ni hisoblang.

A) *1,5* B) *1,25* C) *0,75* D) *0,25*

7. Muntazam to‘rtburchakli kesik piramida asoslarining tomonlari 9 cm va 15 cm. Kesik piramidaning diagonali 18 cm bo‘lsa, uning balandligini (cm) toping.

A) *7* B) *8* C) *6* D) *9*

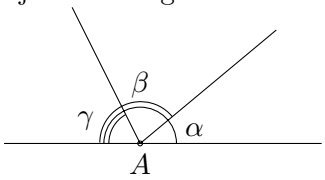
8. $7 \cdot 3^{x+1} - 5^{x+2} = 3^{x+4} - 5^{x+3}$ tenglamaning ildizi quyidagi oraliqlardan qaysi biriga tegishli?

A) $(0; 2]$ B) $(-2; -1]$ C) $(-1; 0]$ D) $(0; 1]$

9. $f(x) = x^2 + bx + c$ funksiyaning nollari 2 va 3 bo‘lsa, c ni toping.

A) -5 B) 6 C) 5 D) -6

10. Yoyiq burchakning A nuqtasidan chiquvchi ikkita nur uni 2:4:3 nisbatdagi burchaklarga ajratadi. Eng katta burchakni toping.



A) *60°* B) *90°* C) *80°* D) *40°*

11. $x^2 + y^2 - 12x + 6y + 9 = 0$ tenglama bilan berilgan aylananing markazini toping.

A) $(3; -6)$ B) $(6; -3)$ C) $(-6; 3)$ D) $(-3; 6)$

12. Uchburchakning 3 va 4 teng bo‘lgan tomonlariga o‘tkazilgan medianalar o‘zaro perpendikulyar bo‘lsa, bu uchburchakning uchinchi tomonini toping.

A) $\sqrt{5}$ B) *2,5* C) $\sqrt{6}$ D) *2,4*

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13. Sakkizta haddan iborat arifmetik progressiyaning toq o‘rindagi hadlari yig‘indisi 168 ga, juft o‘rindagi hadlari yig‘indisi 200 ga teng. Shu progressiyaning oltinchi hadini toping.

A) *50* B) *42* C) *66* D) *58*

14. $\dfrac{\sqrt[3]{x^4}-9}{\sqrt[3]{x^2}-3}-\dfrac{\sqrt[3]{x^2}-4}{\sqrt[3]{x}+2}=7$ tenglamaning ildizlari yig‘indisini toping.

A) *7* B) -1 C) *8* D) *19*

15. Hisoblang:

$$2\cdot\left(\frac{\sin^3 1-\cos^3 1}{\sin 1-\cos 1}-\frac{\sin^2 1+\cos^2 1}{\operatorname{tg} 1+\operatorname{ctg} 1}\right)+1$$

A) $1,25$ B) 2 C) $1,5$ D) 3

16. Aylanada 5 ta har xil nuqta berilgan. Uchlari bu nuqtalarda bo‘lgan jami nechta turli vatar mavjud?

A) *15* B) *6* C) *10* D) *12*

17. Agar a, b, c musbat haqiqiy sonlar uchun $ab=14$ va $bc=6$ bo‘lsa, $a+2b+3c$ eng kichik qiymatini toping.

A) *18* B) *34* C) *20* D) *16*

18. Hisoblang: $\dfrac{\sin 43^\circ+\sin 17^\circ}{2\cos 13^\circ+3\sin 77^\circ}$.

A) $\dfrac{2}{3}$ B) $\dfrac{1}{2}$ C) $\dfrac{1}{5}$ D) 1

19. $\dfrac{\dfrac{2}{x-12}+\dfrac{1}{x+11}}{\dfrac{3}{x+11}-\dfrac{2}{x-12}}=\dfrac{1}{5}$ tenglamaning ildizini toping.

A) $2\dfrac{3}{14}$ B) $-7\dfrac{5}{7}$ C) $18,5$ D) $-6\dfrac{4}{7}$

20. $\dfrac{\sqrt{mn}\cdot\sqrt[4]{m}}{(m+2)\cdot\sqrt[4]{m^{-1}n^2}}-\dfrac{m^2+4}{m^2-4}$ ifodaning $m=6$ va $n=4\sqrt{3}$ bo‘lgandagi qiymatini toping.

A) $-2\sqrt{3}$ B) $\sqrt{3}$ C) $-0,5$ D) 2

21. $\log_3^2(x-1)-2\log_3(x-1)>3$ tengsizlikning barcha haqiqiy yechimlari to‘plamini toping.

A) $\left(1;\dfrac{4}{3}\right)\cup(28;+\infty)$

B) $(28;+\infty)$

C) $\left(1;\dfrac{4}{3}\right)$

D) $\left(-\infty;\dfrac{4}{3}\right)\cup(28;+\infty)$

22. $\left(\left(a^{-2}\right)^{-3}\cdot\left(a^3\right)^{-1}\cdot a^{-2}\right)^3 : a$ ifodaning $a=-\dfrac{1}{6}$ dagi qiymatini toping.

A) $\dfrac{1}{36}$ B) $-\dfrac{1}{6}$ C) 36 D) -6

23. Agar piramida asosining diagonallari soni 44 ta bo‘lsa, bu piramidaning qirralar soni yoqlari sonidan qanchaga ko‘p?

A) *12* B) *11* C) *9* D) *10*

24. Ifodani soddalashtiring:

$$\left(\frac{n^2}{m^3-mn^2}+\frac{1}{n+m}\right):\left(\frac{m-n}{m^2+mn}-\frac{m}{n^2+mn}\right)$$

A) $\dfrac{n}{m-n}$ B) $\dfrac{m}{m+n}$ C) $\dfrac{n}{n-m}$

D) $\dfrac{m}{n-m}$

25. $x\cdot(x+7)^2\geq 5x^2$ tengsizlikning $(1;7)$ oraliqdagi butun yechimlari yig‘indisini toping.

A) *28* B) *21* C) *14* D) *20*

26. $f(x)=\dfrac{x^8\cdot\ln 7x}{8}-\dfrac{x^8}{64}+1$ funksiyaning x_0=1 nuqtadagi hosilasini toping.

A) $2\ln 7$ B) $\dfrac{1}{4}\ln 7$ C) $\ln 7$ D) $\dfrac{1}{2}\ln 7$

27. Magazinda birinchi kuni 76% tarvuz sotildi. Ikkinchi kuni esa qolgan 54 ta tarvuz sotildi. Birinchi kuni nechta tarvuz sotilgan?

A) *168* B) *176* C) *163* D) *171*

28. A aralashmaning bir kilogrammi 12000 so‘m, B aralashmaning bir kilogrammi 18000 so‘m. A va B aralashmalardan mos ravishda 4:1 nisbatda tayyorlangan 1 kg aralashmaning narxini (so‘m) aniqlang.

A) *12800* B) *14200* C) *13200* D) *14800*

29. A, B va C sonli to‘plamlarning elementlari soni mos ravishda 10; 12 va 15 ta. $A\cup B\cup C$ to‘plamning elementlari soni eng kamida necha bo‘la oladi?

A) *22* B) *15* C) *12* D) *27*

30. b^2 ning qanday qiymatlarida $f(x)=x^2+bx+3$ funksiyaning grafigi abssissalar o‘qidan yuqorida joylashadi?

A) $b^2>12$ B) $b^2<12$ C) $b^2>0$
D) $b^2=12$

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Test varianti “Test topshiriqlari to‘plami 2019” asosida shakllantirilgan:

1 – 87-bet 18	7 – 112-bet 43	13 – 30-bet 8	19 – 59-bet 1	25 – 68-bet 18
2 – 45-bet 7	8 – 54-bet 27	14 – 66-bet 75	20 – 29-bet 68	26 – 77-bet 10
3 – 16-bet 63	9 – 72-bet 25	15 – 40-bet 61	21 – 58-bet 32	27 – 19-bet 31
4 – 8-bet 11	10 – 91-bet 11	16 – 123-bet 9	22 – 23-bet 6	28 – 21-bet 46
5 – 114-bet 4	11 – 100-bet 77	17 – 82-bet 47	23 – 107-bet 10	29 – 121-bet 18
6 – 11-bet 11	12 – 102-bet 94	18 – 38-bet 34	24 – 50-bet 51	30 – 76-bet 65

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